

Distribution	$p(x)$	$E[X]$	$Var[X]$
Uniform (Discrete)	$\frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(a-b+1)^2-1}{12}$
Uniform (Continuous)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(a-b)^2}{12}$
Bernoulli	p	p	pq
Binomial	${}^n C_k p^k (1-p)^{n-k}$	np	npq
Geometric	$q^{n-1} p$	$1/p$	q/p^2
Negative Binomial	${}^{n-1} C_{k-1} q^{n-k} p^k$	k/p	kq/p^2
Normal	$\frac{1}{\sigma\sqrt{2\pi}} \cdot e^{\frac{-(x-\mu)^2}{2\sigma^2}}$	μ PMF: $\sum_{x=a}^b x p(x)$ PDF: $\int_a^b x f(x) \cdot dx$	σ^2 PMF: $\sum_{x=a}^b (x-\mu)^2$ PDF: $\int_a^b (x-\mu)^2 f(x) \cdot dx$
Poisson	$\frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ
Exponential	$\lambda \cdot e^{-\lambda k}$	$1/\lambda$	$1/\lambda^2$
Hyper Geometric	$\frac{{}^K C_k {}^{N-K} C_{n-k}}{{}^N C_n}$	$n \frac{K}{N}$	$n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{(N-1)}{(N-1)}$

Inferential Statistics

Tests for One population/ One sample

① z-test

C.I. :-

$$\bar{x} - Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

② t-test

C.I. :-

$$\bar{x} - t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right)$$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

③ chi-squared test

C.I. :-

$$\frac{(n-1) s^2}{\chi^2_{\alpha/2, n-1}} \leq \sigma^2 \leq \frac{(n-1) s^2}{\chi^2_{1-\alpha/2, n-1}}$$

$$\chi^2 = (n-1) \frac{s^2}{\sigma_0^2}$$

④ 2-proportion test

C.I.:-

$$\hat{p} - z_{\alpha/2} \left(\frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \right) \leq p_0 \leq \hat{p} + z_{\alpha/2} \left(\frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \right)$$

$$Z = \frac{\hat{p} - p_0}{\left(\frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}} \right)}$$

Tests for two population /
two samples

① Z-test

C.I.:-

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

② t-test (independent, equal variance)

C.I. :-

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$s_p = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

③ t-test (independent, unequal variance)

C.I. :-

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

④ Paired t-test

C.I. :-

$$\bar{d}_i \pm t_{\alpha/2, n-1} \frac{s_{d_i}}{\sqrt{n}}$$

$$t = \frac{\bar{d}_i - d_0}{\left(\frac{s_{d_i}}{\sqrt{n}} \right)}$$

⑤ F-test

$$F = 8$$

C.I. :-

$$\frac{S_1^2/S_2^2}{F_{1-\alpha/2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2/S_2^2}{F_{\alpha/2}}$$

$$F = \frac{S_1^2}{S_2^2}$$

⑥ Z-proportion test

C.I. :-

$$(\hat{p}_1 - \hat{p}_2) \pm Z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Tests for Three or more states of single variable

① ANOVA

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F-stat.
Between Treatments	$SSB = n \sum_{i=1}^a (\bar{y}_i - \bar{\bar{y}})^2$	$a-1$	$MSB = \frac{SSB}{Dof}$	$F = \frac{MSB}{MSE}$
Error within treatments	$SSE = \sum_{i=1}^a \sum_{j=1}^n (y_{i,j} - \bar{y}_i)^2$	$N-a$	$MSE = \frac{SSE}{Dof}$	
Total	$SST = \sum_{i=1}^a \sum_{j=1}^n (y_{i,j} - \bar{\bar{y}})^2$	$N-1$	$MST = \frac{SST}{Dof}$	

→ compare, F-distribution with, $a-1$, $N-a$ degrees of freedom and get a p-value. against the calculated F-statistic value.

→ if Null Hypothesis is true, $MSB = MSE = MST$

If not, then $MSB > MST > MSE$

② Chi-Squared Test of Independence

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}$$

$$Dof = (r-1) * (c-1)$$

Acceptance Matrix for

Hypothesis tests

		Decision	
		Reject the Null Hypothesis	Failed to Reject the Null Hypothesis
Actual	Null Hypothesis is true	Type I error → Prior to any data collection, Type I error could be as high as alpha (α); and after analysis, it is exactly equal to the p-value.	Correct Decision $(1 - \alpha)$
	Alternate Hypothesis is true	Correct Decision (Power = $1 - \beta$)	Type II error → It is a function of * $\delta = \mu - \mu_0$ * σ * $d = \frac{\delta}{\sigma}$ * n * α {OC curve}

Linear Algebra

Projections:-



projection of b (vector) onto a (line / another vector)

$$p = \hat{x} a$$

$$e = b - p$$

$$\hat{x} = \frac{a^T b}{a^T a}$$

Projection of b (vector) onto subspace $Ax = b$

$$p = A \hat{x}$$

$$e = b - A \hat{x}$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$\therefore p = A (A^T A)^{-1} A^T b$$

Projection Matrix, $P = A (A^T A)^{-1} A^T$

Gram-Schmidt orthogonalization:-

$$a, b, c \rightarrow v_1, v_2, v_3 ; A \Rightarrow Q$$

$$u_1 = a$$

$$u_2 = b - \frac{u_1^T b}{u_1^T u_1} u_1$$

$$u_3 = c - \frac{u_1^T c}{u_1^T u_1} u_1 - \frac{u_2^T c}{u_2^T u_2} u_2$$

$$v_1 = \frac{u_1}{\|u_1\|}$$

$$v_2 = \frac{u_2}{\|u_2\|}$$

$$v_3 = \frac{u_3}{\|u_3\|}$$

$$A = QR$$

$$R = Q^T A$$

$$A \hat{x} = b \Rightarrow A^T A \hat{x} = A^T b$$

$$(QR)^T (QR) \hat{x} = (QR)^T b$$

$$R^T Q^T Q R \hat{x} = R^T Q^T b$$

$$R^T R \hat{x} = R^T Q^T b$$

$$R \hat{x} = Q^T b$$

$$\hat{x} = R^{-1} Q^T b$$

This is equivalent

$$\text{to } \hat{x} = A^{-1} b$$

$$A = QR ; A^{-1} = R^{-1} Q^T$$

Gradient Descent:-

$$x(t+1) = x(t) - \alpha \nabla f(t)$$

Optimization

Unconstrained Optimization:-

$$\min_x f(x)$$

$$J = \nabla f_x = 0$$

$$H = \nabla^2 f_x > 0$$

$$J = \nabla f_x = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

$$H = \nabla^2 f_x = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Eigen values of H should be > 0

Constrained Optimization with 1st order conditions and direction of movement:-

Equality constraint:-

$$\min_x f(x)$$

$$\text{s.t. } h(x) = 0$$

$$L(x, \lambda) = f(x) + \lambda h(x)$$

$$\nabla_x L = 0 \Rightarrow \nabla_x f(x) + \lambda \nabla_x h(x) = 0 \quad (\text{at optimum})$$

$$\nabla_\lambda L = 0 \Rightarrow h(x) = 0$$

$\downarrow$

$$\nabla f = -\lambda \nabla h$$

Inequality constraint :-

$$\begin{aligned} \min_x f(x) \\ \text{s.t. } g(x) \leq 0 \end{aligned}$$

$$L(x, \mu) = f(x) + \mu g(x)$$

$$(\text{at optimum}) \rightarrow \boxed{\nabla f = -\mu \nabla g}$$

$$(\text{inside the boundary}) \rightarrow \boxed{(\nabla f)^T s < 0}$$

$$(\text{on the boundary}) \rightarrow \boxed{(\nabla f)^T s < 0} ; \boxed{(\nabla g)^T s \leq 0}$$

m-equality constraints :-

$$\min_x f(x) \quad - \{x_1, x_2, \dots\}$$

$$\text{s.t. } h_i(x) = 0 \quad - \{h_1(x) = 0, h_2(x) = 0, \dots\}$$

$$\boxed{\nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla h_i(x^*) = 0}$$

$\nabla h_1, \nabla h_2, \dots, \nabla h_m$ are linearly independent

KKT conditions :-

$$\min_{x_i} f(x), \quad \{\text{objective function}\}$$

$$\text{s.t. } g_i(x) \leq 0 \quad \{m\text{-inequality constraints}\}$$

$$\text{s.t. } h_i(x) = 0 \quad \{n\text{-equality constraints}\}$$

For a point x^* to be optimal, there exists multipliers $\lambda_i \geq 0$ (for inequalities) and μ_j (for equalities) such that :-

1. Stationarity :- $\nabla f(x^*) + \sum_{i=1}^m \lambda_i g_i(x^*) + \sum_{j=1}^n \mu_j h_j(x^*) = 0$
 (Gradient of Lagrangian = 0)

2. Primal Feasibility :- $g_i(x^*) \leq 0$; $h_j(x^*) = 0$
 (The solution satisfies original constraints)

3. Dual Feasibility :- $\lambda_i \geq 0$

(Multipliers for inequalities are non-negative)

4. Complementary Slackness :- $\lambda_i \cdot g_i(x^*) = 0 \quad \forall i$

(If a constraint is active, $g_i(x^*) = 0$; then $\lambda_i > 0$)

If a constraint is inactive, $g_i(x^*) < 0$; then $\lambda_i = 0$)

Duality approach :- (Works for linear and Quadratic objective functions)

$$\begin{array}{ll} \min. c^T x & \rightarrow \max. b^T y \\ \text{s.t. } Ax \leq b & \text{s.t. } A^T y \geq c \end{array}$$

For any feasible solution of primal and dual,

(minimization of primal) dual objective $\leq$ Primal objective

(maximization of primal) primal objective $\leq$ Dual objective

Strong Duality

dual obj. = primal obj.

Weak Duality

dual obj. $<$ primal obj.
 (for mini. of primal)

primal obj. $>$ dual obj.
 (for maxi. of primal)

Simplex method:-

- * Works only for linear functions with linear constraints.
- * Slack variables are introduced to convert inequality constraints into equality constraints.